

Extra Space Asia — Recommendation & Booking API

Build chat / web / mobile experiences that quote and reserve self-storage units through Extra Space Asia.

- **Base URL** — `https://backend.extraspace.com.sg`
- **Auth** — X-API-Key: `esa_<prefix>.<secret>` on every request
- **Format** — JSON in/out, UTF-8, ISO dates (YYYY-MM-DD), decimals as numbers (91.01)
- **Status** — current scope is **recommend** → **reserve** → **confirm price**. Lease creation (move-in) is handled by Extra Space Asia operations and is not part of this integration.

Heads up: the recommender is **stateful**. Read [§3 — Identity](#) before writing your first integration. Four context fields glue every step together — get them right and everything just works.

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1 · Quick-start

Five-minute integration test:

```
# 1. Get a quote (3 priced unit slots)
curl -X POST https://backend.extraspace.com.sg/api/recommendations \
-H "X-API-Key: $ESA_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "duration_months": 6,
  "filters": { "location": ["L017"], "size_range": ["30-35"] },
  "context": {
    "channel": "chatbot",
    "request_id": "00000000-0000-0000-0000-000000000001",
    "session_id": "demo-session-1",
    "customer_id": "demo-customer-1"
  }
}'
```

You'll get back **3 slots** (Best Match / Best Alternative / Best Price) with full pricing, per-month breakdowns, and continuation tokens. From there:

- Customer accepts a slot → `POST /api/reservations/reserve` (\$6)
- Show authoritative price → `GET /api/reservations/move-in/cost` (\$7)

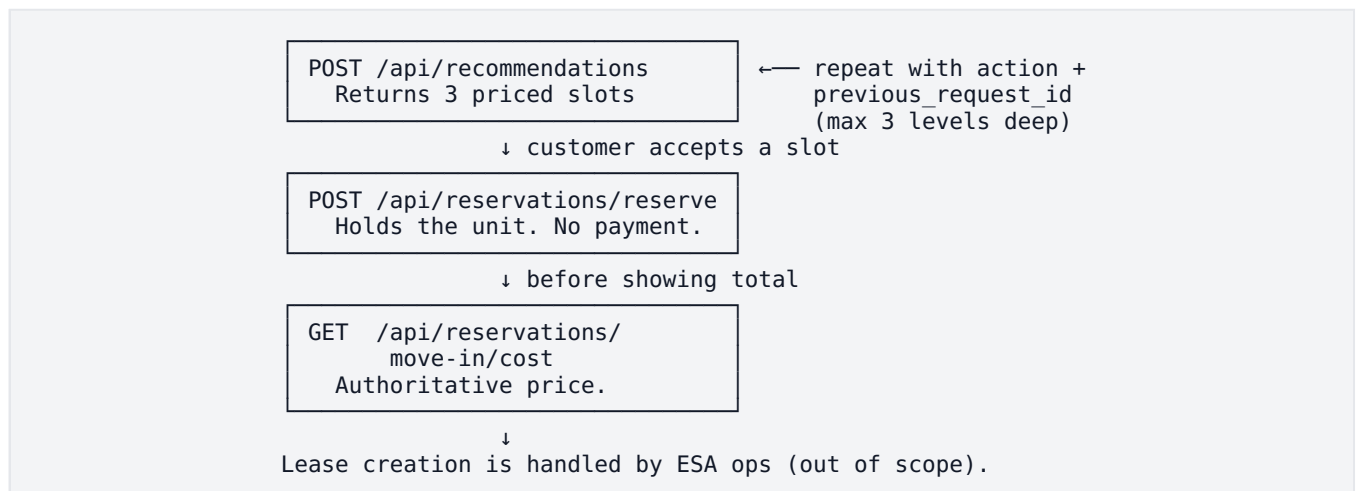
- Customer wants different options → recommend again with `previous_request_id` (§5)

Rate limits

Endpoint	Limit
POST <code>/api/recommendations</code>	120/min
POST <code>/api/reservations/reserve</code>	10/min
GET <code>/api/reservations/move-in/cost</code>	30/min
PUT <code>/api/reservations/<id> (update/cancel)</code>	10/min

HTTP 429 when over, with `retry_after` seconds in the body.

2 · Booking lifecycle



Optional reservation management between **reserve** and lease completion:

Method	Path	Purpose
GET	<code>/api/reservations/<waiting_id></code>	Read state
PUT	<code>/api/reservations/<waiting_id></code>	Modify (move-in date, contact info)
PUT	<code>/api/reservations/<waiting_id>/cancel</code>	Cancel

3 · Identity model

Four context fields are required on every recommend call. They tie every step of one customer's flow together — from first quote through follow-up turns to the held reservation.

Field	Format	Lifetime	Source
<code>channel</code>	enum: chatbot web api admin	per call	hardcoded by your integration
<code>request_id</code>	string ≤ 64 chars (UUID v4 recommended)	unique per turn	bot mints fresh per call
<code>session_id</code>	string ≤ 64 chars	per conversation — same on every recommend turn AND on / reserve	bot mints once at chat start
<code>customer_id</code>	string ≤ 64 chars	per customer lifetime	your channel's stable user id (or stable surrogate for anonymous flows)

Plus one optional field for follow-up turns:

Field	When to send
previous_request_id	The request_id of the prior recommend turn. Tells the engine "this is a continuation". See §5.

Validation rules - All four core fields are mandatory; missing/empty/whitespace-only → HTTP 400. - Length cap: 64 chars per field. - Case-sensitive. - Server returns next_turn.request_id + next_turn.session_id in the response so you can confirm what got logged.

4 • POST /api/recommendations

Quote engine. Returns **up to 3 priced slots** matching the customer's intent.

Minimal request

```
curl -X POST https://backend.extraspace.com.sg/api/recommendations \
-H "X-API-Key: $ESA_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "duration_months": 6,
  "filters": { "location": ["L017"] },
  "context": {
    "channel": "chatbot", "request_id": "<uuid>",
    "session_id": "<uuid>", "customer_id": "user-42"
  }
}'
```

Full request body

```
{
  "mode": "recommendation",
  "duration_months": 6,
  "filters": {
    "location": ["L017"],
    "unit_type": ["W", "L"],
    "climate_type": ["A", "AD"],
    "size_range": ["30-35"],
    "unit_id": [107197],
    "concession_id": 11872,
    "plan_id": [10],
    "coupon_code": "PROM0123"
  },
  "context": {
    "channel": "chatbot",
    "request_id": "550e8400-e29b-41d4-a716-446655440000",
    "session_id": "sess-conv-7",
    "customer_id": "user_xyz_123",
    "previous_request_id": "...",
    "picked_slot": 1,
    "action": "more_like_this"
  },
  "constraints": {
    "max_distance_km": 50,
    "exclude_unit_ids": [107197]
  }
}
```

Field	Required	Notes
mode	optional	recommendation (default) or quote (single unit; requires filters.unit_id)
duration_months	yes	integer 1-120
filters.location	yes	array of site codes
filters.unit_type	optional	W/WN/L/U/M/LL (see §10)
filters.climate_type	optional	A/AD/NC/RF (see §10)
filters.size_range	optional	sqft buckets like "30-35"
filters.unit_id	optional	array of integers — pin to specific unit(s)
filters.plan_id	optional	array of integers — cross-site brand filter (one plan covers multiple per-site concessions)
filters.concession_id	optional	single integer — pins a specific (unit, plan, concession) tuple
filters.coupon_code	optional	string — gates coupon-only plans
context.*	mostly yes	see §3
constraints.max_distance_km	optional	overrides default neighbour radius (50 km)
constraints.exclude_unit_ids	optional	up to 200 ids — won't be shown

Sample response

```

{
  "mode": "recommendation",
  "request_id": "550e8400-e29b-41d4-a716-446655440000",
  "served_at": "2026-05-04T01:23:45Z",

  "stats": {
    "recommendation_level": 1,
    "candidates_pool_size": 14,
    "relax_strategy_used": "none",
    "saturation_signal": false
  },

  "slots": [
    {
      "slot": 1, "label": "Best Match",
      "unit_id": 107197, "facility": "L017",
      "unit_type": "W", "climate_type": "AD",
      "size_range": "30-35", "size_sqft": 32.5,
      "plan_id": 10, "plan name": "Moving Season SG",
      "concession_id": 11872, "concession_name": "SS-TAC-Move-R-30%",
      "discount_summary": "30% off every month",

      "match_flags": {
        "alternative_strategy": null,
        "relaxed_dims": [],
        "distance_km": 0,
        "travel_warning": false
      },

      "terms": {
        "discount_perpetual": true,
        "prepayment_months": 6,
        "post_prepay_uplift_pct": 5.0,
        "lock_in_months": 0,
        "promo_valid_until": "2026-12-31"
      },

      "pricing": {
        "first_month_total": 238.71,
        "total_due_at_movein": 693.76,
        "total_contract": 693.76,
        "monthly_average": 115.63,
        "rate_during_prepay": 80.50,
        "rate_after_prepay": 84.52,
        "rate_change_date": "2026-11-15",
        "monthly_all_in_during_prepay": 91.01,
        "monthly_all_in_after_prepay": 95.40,
        "monthly_insurance_premium": 3.00,
        "monthly_insurance_tax": 0.27,
        "breakdown": [
          { "month_index": 1, "billing_date": "2026-05-15",
            "rent": 115.00, "discount": 34.50, "rent_proration_factor": 1.0,
            "insurance": 3.00, "deposit": 296.00, "admin_fee": 30.00,
            "rent_tax": 7.25, "insurance_tax": 0.27, "total": 238.71 },
          { "month_index": 2, "billing_date": "2026-06-15",
            "rent": 115.00, "discount": 34.50, "insurance": 3.00,
            "deposit": 0, "admin_fee": 0, "total": 91.01 }
          /* ... one entry per month of duration_months ... */
        ]
      },

      "insurance": {
        "selected": { "id": 1, "coverage": 1000, "premium": 3.00 },
        "options": [
          { "id": 1, "coverage": 1000, "premium": 3.00 },
          { "id": 2, "coverage": 2000, "premium": 7.00 }
        ],
        "min_required": 1000
      },

      "customer_disclosure": {
        "fine_print": [
          "Pay $693.76 today to lock 6 months at $91.01/month all-in.",
          "After 2026-11-15, monthly adjusts to $95.40 all-in (rent +5% adjustment).",

```

```

    "Insurance is changeable – see insurance.options for higher coverage tiers."
  ]
},
{
  "slot": 2, "label": "Best Alternative",
  "match_flags": { "alternative_strategy": "same_site_2nd" },
  "...": "same shape as slot 1"
},
{
  "slot": 3, "label": "Best Price",
  "match_flags": { "relaxed_dims": ["unit_type"], "savings_pct": 18.3 },
  "...": "same shape – strictly cheaper than slot 1"
}
],

"next_turn": {
  "previous_request_id": "550e8400-e29b-41d4-a716-446655440000",
  "session_id": "sess-conv-7",
  "supported_actions": ["more_like_this", "bigger_size", "smaller_size",
    "expand_locations", "different_type", "different_duration"],
  "next_level_allowed": true
},

"reserve_template": {
  "endpoint": "POST /api/reservations/reserve",
  "site_code": "L017",
  "unit_id": 107197,
  "concession_id": 11872,
  "plan_id": 10
}
}

```

Slot semantics

Slot Label	What it is	Distinguished by match_flags
1 Best Match	Cheapest unit matching every filter	alternative_strategy: null, relaxed_dims: []
2 Best Alternative	Most convenient alternative	alternative_strategy: same_site_2nd / neighbour_close / neighbour_far
3 Best Price	Strictly cheaper unit at the same site	relaxed_dims: which dim was dropped; savings_pct

When the strict-filter pool is empty, the engine runs **pool rescue** — it relaxes dimensions until candidates appear. Surfaced as `stats.saturation_signal: true + stats.pool_rescue_step`. Each slot's `match_flags.relaxed_dims` shows what was relaxed for that specific slot.

The response is guaranteed to contain **≥ 2 slots in normal operation**. Slot 3 is best-effort; null if no strictly-cheaper unit exists.

mode=quote — single-unit pricing

For "what does unit X cost?" — pass `mode: "quote"` plus `filters.unit_id: [N]`. Returns one slot keyed to that unit. Pricing is **cent-identical** to what `mode=recommendation` returned for the same unit.

```

{
  "mode": "quote",
  "duration_months": 6,
  "filters": { "location": ["L017"], "unit_id": [107197] },
  "context": { "channel": "chatbot", "request_id": "...", "session_id": "...", "customer_id": "..." }
}

```

5 • Multi-turn search

The recommender is **hierarchical, max 3 levels deep** per booking flow. Pass `previous_request_id` + action to continue from a prior turn.

Level 1	Initial recommend	— no <code>previous_request_id</code>
Level 2	Continuation off L1	— <code>previous_request_id</code> = L1. <code>request_id</code>
Level 3	Continuation off L2	— <code>previous_request_id</code> = L2. <code>request_id</code>
Level 4	REJECTED with HTTP 400 — customer must accept a slot or pivot to a fresh L1	

`stats.recommendation_level` (1, 2, or 3) tells you where you are.

`next_turn.next_level_allowed` flips to false on level-3 — that's your signal to stop chaining.

Action vocabulary

All actions require `previous_request_id`.

action	What the engine does
<code>more_like_this</code> (paired with <code>picked_slot=1\ 2\ 3</code>)	Slot-specific tightening. Slots 1/3 → <code>size_range</code> ±1 bucket. Slot 2 → next-nearest site.
<code>bigger_size</code>	Shift <code>size_range</code> ONE bucket up (e.g. 30-35 → 35-40)
<code>smaller_size</code>	Shift <code>size_range</code> ONE bucket down
<code>expand_locations</code>	Add nearest neighbour sites (within <code>max_distance_km</code>) to <code>filters.location</code>
<code>different_type</code>	Drop the <code>unit_type</code> filter
<code>different_duration</code>	Analytics signal — bot also changes <code>duration_months</code> so the engine re-quotes against the new length

Decision tree

Customer says...	Bot's next call
"I'll take slot 1 / slot 2"	Skip recommend → POST <code>/reserve</code>
"More like that one" / "Anything similar to slot 2?"	recommend with <code>previous_request_id</code> + <code>picked_slot</code> + action: " <code>more_like_this</code> "
"Other options?" / "What else?"	recommend with <code>previous_request_id</code> alone — auto-excludes shown units
"Something bigger" / "Something smaller"	recommend with <code>previous_request_id</code> + action: " <code>bigger_size</code> " (or <code>smaller_size</code>)
"What about nearby locations?"	recommend with <code>previous_request_id</code> + action: " <code>expand_locations</code> "
"Different unit type?"	recommend with <code>previous_request_id</code> + action: " <code>different_type</code> "
"What if I lease for 12 months?"	recommend with new <code>duration_months</code> + action: " <code>different_duration</code> "
"Actually I want something completely different"	Fresh L1: new filters, new <code>request_id</code> , keep <code>session_id</code> + <code>customer_id</code> , drop <code>previous_request_id/picked_slot/action</code>

6 • POST `/api/reservations/reserve`

Holds the unit and creates a tenant + reservation record. **No payment is taken.**

Minimal request

```
curl -X POST https://backend.extraspace.com.sg/api/reservations/reserve \
-H "X-API-Key: $ESA_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "site_code": "L017", "unit_id": 107197, "concession_id": 11872,
  "first_name": "Jane", "last_name": "Tan", "phone": "+6591234567",
  "needed_date": "2026-05-15",
  "session_id": "sess-conv-7",
  "customer_id": "user_xyz_123"
}'
```

Body fields

Field	Required Notes	
site_code	yes	from the picked slot
unit_id	yes	from the picked slot
concession_id	yes	from the picked slot. 0 = standard rate — pass 0, never null
plan_id	optional	from the picked slot; attribution signal
first_name, last_name, phone	yes	tenant identity
email, mobile, address, city, postal_code, country	optional	
needed_date	optional	YYYY-MM-DD, default = tomorrow
comment, quoted_rate, source_name	optional	
session_id, customer_id	yes	must match the recommend call
previous_request_id	optional	the picked slot's recommend request_id — strongest attribution

Response

```
{
  "success": true,
  "site_code": "L017",
  "unit_id": 107197,
  "tenant_id": "1109544",
  "waiting_id": "848757",
  "global_waiting_num": "809363726",
  "message": "Reservation created"
}
```

Hold on to `waiting_id` and `tenant_id` — both identify the held reservation for any subsequent read / modify / cancel call.

7 • GET /api/reservations/move-in/cost

Authoritative price for the held reservation. Call this **right before showing the customer the total**.

Minimal request

```
curl -G "https://backend.extraspace.com.sg/api/reservations/move-in/cost" \
-H "X-API-Key: $ESA_API_KEY" \
--data-urlencode "site_code=L017" \
--data-urlencode "unit_id=107197" \
--data-urlencode "concession_id=11872" \
--data-urlencode "insurance_id=1" \
--data-urlencode "move_in_date=2026-05-15" \
--data-urlencode "waiting_id=848757" \
--data-urlencode "variant=reservation"
```

Query parameters

Param	Required	Notes
site_code	yes	
unit_id	yes	
concession_id	optional	same value used on /reserve; default 0
insurance_id	optional	the slot's selected insurance, or 0 for none
move_in_date	optional	default = tomorrow
variant	optional	standard (default) reservation 28day push_rate. Use reservation for a held reservation.
waiting_id	required when variant=reservation	from /reserve
promo_id	optional	only with variant=reservation

Response

```
{
  "site_code": "L017", "unit_id": 107197, "move_in_date": "2026-05-15",
  "tenant_rate": 80.50, "discount": 24.15,
  "total": 238.71,
  "charges": [
    { "description": "First Monthly Rent Fee", "amount": 80.50, "tax": 7.25, "total": 87.75 },
    { "description": "Administrative Fee", "amount": 30.00, "tax": 2.70, "total": 32.70 },
    { "description": "Security Deposit", "amount": 296.00, "tax": 0.00, "total": 296.00 },
    { "description": "First Month Insurance", "amount": 3.00, "tax": 0.27, "total": 3.27 }
  ]
}
```

The total field is the authoritative amount.

8 · Reservation lifecycle

For reading or modifying a held reservation.

Read

```
curl "https://backend.extraspace.com.sg/api/reservations/848757?site_code=L017" \
-H "X-API-Key: $ESA_API_KEY"
```

Returns the reservation's current state.

Modify

```
curl -X PUT https://backend.extraspace.com.sg/api/reservations/848757 \
-H "X-API-Key: $ESA_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "site_code": "L017",
  "needed_date": "2026-05-20",
  "first_name": "Jane",
  "phone": "+6591234567",
  "comment": "Customer changed move-in date"
}'
```

Only fields you include in the body are written. Returns `{"success": true, "waiting_id": ..., "message": "Reservation updated"}`.

Cancel

```
curl -X PUT https://backend.extraspace.com.sg/api/reservations/848757/cancel \
-H "X-API-Key: $ESA_API_KEY" \
-H "Content-Type: application/json" \
-d '{"site_code": "L017"}'
```

Idempotent — cancelling an already-cancelled reservation returns success.

9 • End-to-end

A 3-turn chat ending in a held reservation with confirmed price.

```
// — Turn 1 – fresh recommend —————
POST /api/recommendations
{
  "duration_months": 6,
  "filters": { "location": ["L017"], "size_range": ["30-35"] },
  "context": {
    "channel": "chatbot",
    "request_id": "req-aaaa-1111",
    "session_id": "sess-conv-7",
    "customer_id": "user_xyz_123"
  }
}
// → 3 slots · stats.recommendation_level = 1 · next_level_allowed = true

// — Turn 2 – "show me something bigger" —————
POST /api/recommendations
{
  "duration_months": 6,
  "filters": { "location": ["L017"], "size_range": ["30-35"] },
  "context": {
    "channel": "chatbot",
    "request_id": "req-bbbb-2222",
    "session_id": "sess-conv-7", // unchanged
    "customer_id": "user_xyz_123", // unchanged
    "previous_request_id": "req-aaaa-1111",
    "action": "bigger_size"
  }
}
// → 3 fresh slots in 35-40 range · stats.recommendation_level = 2

// — Customer picks slot 1 – hold the unit —————
POST /api/reservations/reserve
{
  "site_code": "L017",
  "unit_id": 107298,
  "concession_id": 11872,
  "first_name": "Jane", "last_name": "Tan", "phone": "+6591234567",
  "email": "jane@example.com",
  "needed_date": "2026-05-15",
  "session_id": "sess-conv-7",
  "customer_id": "user_xyz_123",
  "previous_request_id": "req-bbbb-2222"
}
// → tenant_id = 1109544, waiting_id = 848757

// — Confirm authoritative price right before showing the total —————
GET /api/reservations/move-in/cost
  ?site_code=L017
  &unit_id=107298
  &concession_id=11872
  &insurance_id=1
  &move_in_date=2026-05-15
  &waiting_id=848757
  &variant=reservation
// → { total: 238.71, charges: [...] }
```

10 • Reference

Unit type codes

Code	Meaning
------	---------

W	Walk-in (full-size storage room)
WN	Wine storage
L	Locker (smaller, rack-style)
U	Upper-position locker
M	Mid-position locker
LL	Lower-position locker

Climate codes

Code	Meaning
------	---------

A	Air-conditioned
AD	Air-conditioned + dehumidified
NC	Non-climate
RF	Refrigerated (specialised)

Channel codes

Code	Meaning
------	---------

chatbot	Conversational bot integration
web	Direct web booking
api	Generic API caller
admin	Admin tooling

Common error responses

HTTP	Where	What it means
400	any endpoint	Invalid input — body has <code>error</code> field with the specific failure
401	any endpoint	Missing / wrong <code>X-API-Key</code>
403	any endpoint	API key doesn't have the required scope
400	/ recommendations	recommendation chain exceeds max depth (3) — drop <code>previous_request_id</code> and start a fresh L1
200 + success: false	/reserve	Unit got rented or unavailable. Body's message explains. Re-recommend with <code>constraints.exclude_unit_ids: [<that unit_id>]</code> .
429	any endpoint	Rate limit exceeded; <code>retry_after</code> seconds in body

11 • Best practices

1. **Send all four context IDs on every recommend call.** Keep `session_id` + `customer_id` consistent across `recommend` → `reserve`.
2. **Mint a fresh request_id per turn.** Don't reuse it; it's the link key for `previous_request_id` on the next turn.
3. **Re-fetch /move-in/cost immediately before showing the total** to the customer. Pricing can shift if a session is long-running.
4. **Treat HTTP 200 + success: false as a failure** (most often "unit no longer available"). Re-recommend with the failed `unit_id` in `constraints.exclude_unit_ids`.
5. **Stop chaining at level 3.** Watch `next_turn.next_level_allowed` — `false` means encourage the customer to pick a slot or rephrase.
6. **concession_id=0 = standard rate.** Pass 0, never null, when the picked slot has no discount.

7. **Don't share API keys.** One key per integration; rotate via Extra Space Asia ops if compromised.
-

12 • Support

- **Integration questions** — contact your Extra Space Asia integration manager.
- **Liveness check** — GET `/api/health` returns `{"status": "ok"}` when the API is up.
- **API key management** — `https://backend.extraspace.com.sg/api-keys/` (auth required).